

ExpMD: an Explainable Framework for Traffic Identification Based on Multi-Domain Features

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Abstract—Network traffic identification has a significant impact on the QoS (Quality of Service) of network. However, there are challenges in performing network traffic identification, including underutilization of information and weak of interpretability. To address these issues, in this paper, we propose an Explainable framework based on Multi-Domain features for network traffic identification, named ExpMD. The network information is divided into four feature domains: metadata domain, byte-distribution (BD) domain, textual domain, and temporal domain. On these feature domains, we construct four identification models independently, thereby achieving the extraction and utilization of multi-domain information of network flows. Subsequently, efficient identification of network traffic can be achieved through the ensemble of models from four domains. In addition, we employ post-hoc explainable approaches to attribute features across multiple feature domains, thus mitigating the black-box issue of network traffic identification. We run a set of experiments in the identification tasks of application network traffic. The extensive evaluation demonstrates that our framework not only achieves an accuracy of 97.5%, but also generates high-fidelity, sparse, complete, and stable explanation results for network flows. Furthermore, each sub-model performed at a state-of-the-art effect within their feature domains, respectively.

Index Terms—Multi-Domain Features, Network Traffic Analysis, Identification, Ensemble, Explanation

I. INTRODUCTION

Network traffic identification is important in network management and cyber security, and has a significant impact on the QoS (Quality of Service) of network. Over recent decades, with the application of encryption technology, traditional Deep Packet Inspection (DPI) methods fail in encrypted traffic analysis. Currently, encrypted traffic and non-encrypted traffic coexist in network environment, which brings great challenges to the network traffic analysis.

Network traffic identification is the process of identifying the source, type, behavior, and quality of network flows by analyzing their characteristics. A network flow, which is a packet sequence defined by 5-tuple (Source IP, Destination IP, Source Port, Destination Port, and Protocol), is generally the basic analysis unit. At present, machine-learning-based methods [1]–[6] have been widely applied for network traffic identification. In recent years, with the development of deep learning, many end-to-end deep network approaches have also been applied in traffic identification tasks [7]–[12] and achieved good results. However, existing methods still have two main problems:

- **Underutilization of Information:** The logs, loads, and interaction processes of network flows reflect the information of network flows from different perspectives, all of which can be applied for network traffic identification. Most existing methods use network flow information from only a single domain, such as transmission load [8], [10], [12]–[15] or packet length sequence [3], [5], [6], [9], [11]. Consequently, these methods do not sufficiently extract the information of network flows, thus constraining their identification performance to some extent.
- **Weak of Interpretability:** As the complexity of models increases, their interpretability becomes lower, giving rise to the issue of "black-box", whose details are hard to infer [16]. The black-box model can only output predictions without explanations making it less trustworthy [17]. On some tasks, such as intrusion detection [18], [19], detection results are often unconvincing without explanation.

Furthermore, there are also some works [1], [4], [20], [21] for specific types of network traffic, such as TLS. Because their feature extraction is based on specific protocol, these methods cannot be universally applied for network traffic identification.

To address these issues, in this paper, we propose an **Explainable** framework that leverages information from **Multi-Domain** features of network flows for network traffic identification, named ExpMD. The main idea is to make full use of the information of network flows from various domains, enabling efficient identification and feature attribution of network traffic. The network information is divided into four feature domains: metadata domain, byte-distribution (BD) domain, textual domain, and temporal domain. On the four feature domains, we construct four identification models independently, thereby achieving the extraction and utilization of multi-domain information of network flows. Subsequently, efficient identification of network traffic can be achieved through the ensemble of multi-domain models. In addition, we employ explainable methods to attribute features across multiple feature domains to improve the credibility of the identification result.

ExpMD is a universal network traffic identification framework that has following advantages. First, ExpMD achieves a comprehensive representation of network flows through the sufficient extraction of multi-domain information. In the process of multi-domain ensemble, the information from different feature domains can complement and reinforce each other. And even when some feature domain information is missing, the framework is still able to work. Moreover, the representation of

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network flows is protocol-agnostic, so ExpMD can be applied to various types of network traffic, regardless of whether it is encrypted or not. In addition, we also attribute the identification results to different feature domains, effectively alleviating the "black box" issue.

The key contributions of this paper are as follows:

- 1) **We propose a novel idea for multi-domain representation of network flows, dividing the information into metadata domain, byte-distribution domain, textual domain and temporal domain.** We designed a universal framework for network traffic analysis, named ExpMD. After exhaustively extracting the multi-domain information of network flows, the network traffic identification can be realized based on a soft voting ensemble approach.
- 2) **On the four feature domains, we develop four identification models: Meta-RF, BD-boost, Text-Net, and PS-Net.** Among them, we incorporate a byte embedding layer and residual network structure in Text-Net, and introduce a contrastive learning task during the training process of PS-Net. These designs significantly improve the model's capacity for representation learning in their respective feature domains.
- 3) **We employ post-hoc explainable approaches to attribute the identification results on the four feature domains, respectively.** Consequently, it is able to effectively mitigate the "black-box" issue of models.
- 4) We conducted a set of experiments to demonstrate the effectiveness of our approach. ExpMD achieved a identification accuracy of 97.5% in the application network traffic classification tasks. Each sub-model performed at a state-of-the-art effect within their feature domains, respectively. Furthermore, ExpMD can also generate high-fidelity, sparse, complete, and stable explanation results for network flows.

The remainder of this paper is organized as follows. We give a review of related work in Section II. Section III shows the overview of the proposed ExpMD framework, and Section IV introduces its design details. Then, we evaluate the effectiveness of the proposed approaches through a set of experiments in section V. Discussions about our approach are presented in Section VI. Finally, Section VII concludes the paper and discusses future works.

II. RELATED WORK

According to the utilized feature domain, related works can be mainly divided into textual-domain-based methods and temporal-domain-based methods.

The textual domain information of network flows is mainly contained in the load of application layer. Convolutional Neural Network (CNN) is most often applied in the feature extraction of the textual information [7], [8], [10], [13], [14]. Wang *et al.* [7], [13] first presented a new taxonomy of traffic classification from an artificial intelligence perspective. Lotfollahi *et al.* [8] presented Deep Packet, a framework that automatically extracts features from network traffic. Zhang *et al.* [10] proposed an autonomous model updating framework

that can filter the packets of unknown classes and cluster them to the corresponding classes. Zheng *et al.* [14] tried to learn representative features from the raw flows and then classifies them in a unified meta-learning framework. Besides CNN, other network structure can also be applied in the feature extraction of the textual information [12], [15], [22], [23]. Li *et al.* [22] applied Recurrent Neural Network (RNN) to directly map network datagrams to application protocols. Mao *et al.* [23] and Xiao *et al.* [15] employed hierarchical attention networks to jointly learn the high-level presentation from byte sequence and side-channel features. Lin *et al.* [12] proposed ET-BERT, which pre-trains deep contextual datagram-level traffic representations from large-scale unlabeled data.

In temporal domain, packet-length sequence is the most widely use network flow information for network traffic identification [2], [3], [5], [6], [9], [11], [16]. Taylor *et al.* [2], [3] presented AppScanner, which only applies side-channel data to build fingerprints for Android apps. FlowLens [5] proposed by Barradas *et al.* collects features of packet distributions at line speed and classifies flows directly on the switches. Xu *et al.* [6] introduced path signature into encrypted traffic classification based on packet length information. As for deep-learning-based methods, Liu *et al.* [9] designed an RNN-based end-to-end model named FS-Net, which jointly learns the representative features from the raw flow sequences and classifies them together. Shen *et al.* [11] constructed Traffic Interaction Graph (TIG) as an information-rich representation of encrypted flows based on the packet length and packet direction. Song *et al.* [16] presented an incremental and interpretable RNN model for encrypted traffic classification. As for TLS/SSL encrypted traffic, TLS/SSL message sequence is also widely applied for identification [1], [4], [20], [21], [24].

In a word, most existing methods use network flow information from only a single domain (textual domain or temporal domain). They have not make full use of the information of network flows from multi-domain, which limits the effectiveness of network traffic identification.

III. OVERVIEW

Serving as the fundamental unit of network behavior, network flows are integral in the analysis of most network traffic. Network flow is a sequence of packets defined by a 5-tuple (srcIP, srcPort, dstIP, dstPort, Protocol), encompassing both upload and download flows. Most network traffic analysis works are achieved by flow-based classification. From various perspectives such as transmission payload and the transmission process, a multitude of features can be extracted to describe network flows comprehensively. Specifically, network flows comprise four distinct feature domains:

- **Metadata Domain:** The metadata domain typically involves information directly associated with the network connection itself, reflecting the fundamental holistic attributes of network flows. These features include 5-tuple, duration, number of packets, number of bytes, and more, which can be used for network traffic analysis. Network logs (such as NetFlow) typically record this information.

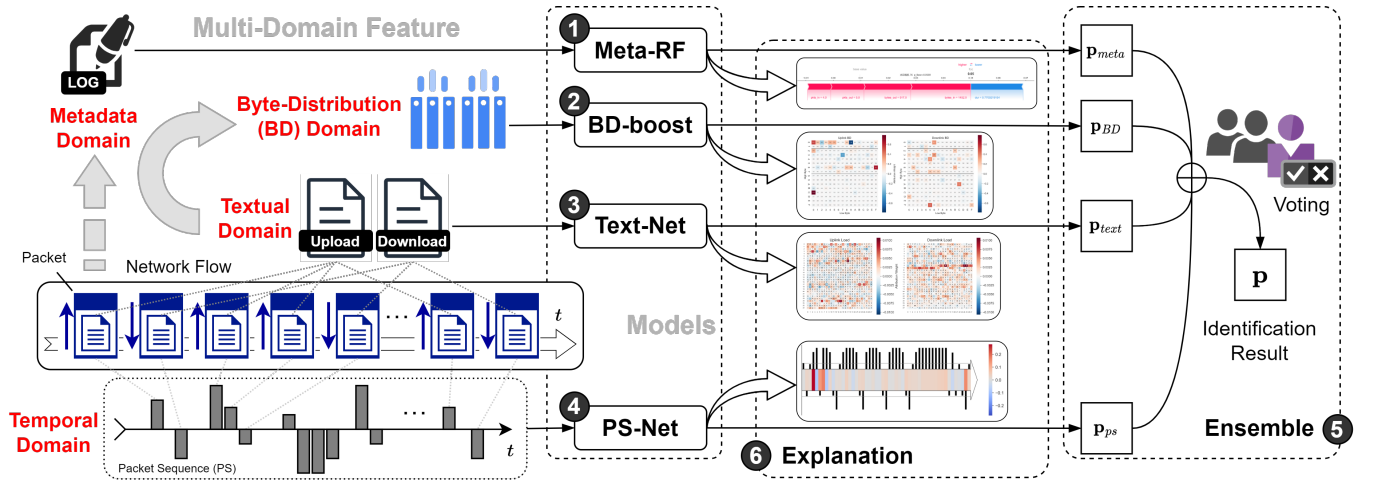


Fig. 1. In ExpMD, there are 4 feature domains for network flows: metadata domain, BD domain, textual domain, and temporal domain, corresponding to 4 sub-models, namely Meta-RF, BD-boost, Text-Net, and PS-Net. They not only provide identification results but also perform feature attribution within their feature domains. By soft voting ensemble from the four sub-models, the final identification result can be obtained.

- **Byte-Distribution (BD) Domain:** This type of feature primarily originates from the distribution of different bytes within the transmission payload of network connections. BD provides information about the encoding of the data [25], [26]. Through BD analysis, we can discern information such as whether the payload is encrypted and the type of file being transmitted to a certain extent [27].
- **Textual Domain:** Textual domain features primarily stem from the transport layer payload of network flows. In the case of plaintext traffic, textual domain features typically encompass features related to the application layer protocol and its payload. For encrypted traffic, such as TLS, textual domain features mainly involve the initial limited plaintext information exchange, which can be exploited for encrypted traffic identification. Furthermore, some studies [7], [13], [25], [26] have indicated that the initial few packets of network flows play a more significant role in network traffic analysis, especially for encrypted traffic. Hence, textual domain features are commonly extracted from the first N bytes of the payload.
- **Temporal Domain:** Temporal domain features are mainly from the packet transmission patterns within network flows. Within network flows, the interaction process of packets generates sequential information (such as packet length sequences), in which temporal features can be mined. These features can reflect information about network behavior and can also be used for analysis. For encrypted traffic analysis, temporal domain features are more important as the plaintext information is limited.

As shown in Fig. 1, ExpMD is designed for explainable traffic identification that utilizes multi-domain information. For the raw network traffic, we extract the features of network flows from the metadata domain, byte-distribution domain, textual domain, and temporal domain, respectively. Specifically, we record log information, the BD of the payload, the byte

stream of the payload, and the packet-length sequences for each network flow. In ExpMD, we construct different models for each feature domain, *i.e.*, Meta-RF for metadata domain, BD-boost for BD domain, Text-Net for textual domain, and PS-Net for temporal domain.

For an unknown network flow, we first extract the information from its various feature domains, and then input them into their corresponding single domain model respectively. These models can give identification results on their feature domains. After that, we combine these identification results by soft voting mechanism to obtain the final identification result. In addition, we attribute the identification result in different feature domains to enhance the interpretability of models, thereby improving the confidence of the identification result.

IV. DESIGN DETAILS

In this section, we introduce the design details of ExpMD, including four single-domain identification sub-models, ensemble method, and feature attribution.

A. Meta-RF for Metadata Domain

In ExpMD, Meta-RF is an identification model based on Random Forest (RF) [28] and utilizes the metadata information of network flow for identification. The features extracted by Meta-RF include: number of up/down packets, total number of up/down bytes, and flow duration. These features are usually recorded directly or can be calculated from the logs.

Random forest [28] combines the power of multiple decision trees to make more accurate predictions. It is an ensemble learning approach based on Bagging. It works by creating a collection of decision trees during training, where each decision tree is built using a random subset of the dataset and a random set of features. This randomness helps prevent overfitting, a common issue with individual decision trees. To make a final prediction, random forest aggregates the results

from all the individual trees. For an unknown sample $\mathbf{x}$, the prediction result $\hat{y}$ is expressed as follows:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^B f_i(\mathbf{x}), \quad (1)$$

where $f_i(\cdot)$ is i -th decision tree of the Random Forest.

B. BD-boost for BD Domain

In ExpMD, BD-boost is an identification model based on XGBoost (eXtreme Gradient Boosting) [29] and utilizes the byte-distribution (BD) information of network flow for identification. We divide the network flow load into two parts: the upload and the download. The proportion of different bytes is counted for each of the two payloads to generate two 256-dimensional vectors. Then, we concatenate these two BD vectors to form the input feature vector (512 dimensions).

XGBoost [29] is an optimized distributed gradient boosting algorithm. It is an ensemble learning method that combines multiple decision trees to improve accuracy and reduce overfitting. XGBoost is designed to efficiently, flexibly and conveniently implement Gradient Boosting Decision Tree (GBDT) [30]. In the m -th iterations, the m -th decision tree $T(\cdot; \Theta_m)$ is applied to fit the residual of the fitting function $f_{m-1}(\mathbf{x})$. Then, the fitting function in the m -th iterations can be expressed as follows:

$$f_m(\mathbf{x}) = f_{m-1}(\mathbf{x}) + T(\cdot; \Theta_m). \quad (2)$$

After M iterations, the prediction result $\hat{y}$ is the sum of all decision tree results, that is,

$$\hat{y} = \sum_{m=1}^M T(\mathbf{x}; \Theta_m) \quad (3)$$

Compared with traditional GBDT, there are many salient features in XGBoost, including clever penalization of trees, a proportional shrinking of leaf nodes, Newton Boosting and extra randomization parameter. To avoid overfitting, XGBoost expands the objective function to a second-order form using Taylor's formula, and also introduces a regularization term that represents the complexity of the model.

C. Text-Net for Textual Domain

In ExpMD, Text-Net is an identification model based on one-dimensional convolutional neural networks (1D-CNN), which can automatically extract textual domain features from network stream payloads through representation learning. The detailed structure of Text-Net is shown in Fig. 2.

For a network flow, we intercept the first 784 bytes of the upload and the download, respectively, and fill the parts that are less than 784 bytes with [PAD]. This process can not only effectively cover the textual information of the first few packets considered important in the network flow, but also avoid the difficulty of extracting textual domain features from the elephant flow.

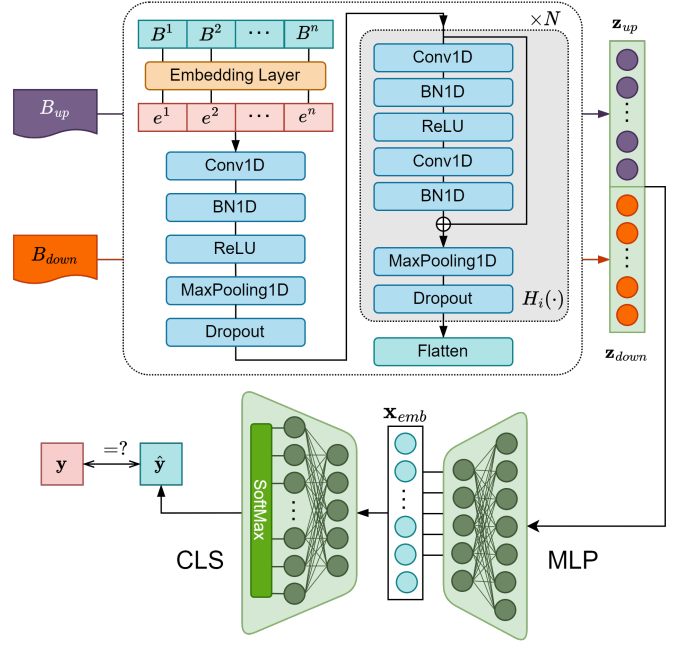


Fig. 2. The network structure of Text-Net.

In the load of network flow, the value proximity of different bytes does not reflect their actual meaning similarity. For instance, non-adjacent bytes may potentially exhibit more meaning similarity than adjacent bytes in some cases. Therefore, employing the values of raw bytes as inputs would introduce misleading information to the model, thereby affecting the effectiveness of representation learning. Byte embedding techniques can effectively mitigate this issue. Consequently, we transform the byte stream of input load $\mathbf{B} = [B_1, B_2, \dots, B_n]$ into a byte embedding sequence $\mathbf{E} = [e_1, e_2, \dots, e_n]$ through the embedding layer, firstly.

As illustrated in Fig. 2, with the first "convolution-pooling" structure, the local information of the sequence $\mathbf{x}_1$ can be preliminarily extracted, that is, $\mathbf{x}_1 = \text{CNN}(\mathbf{E})$. Then, the extraction of global sequence features can be realized by multi-layer blocks. Among them, the design of the block apply the residual network structure, which can avoid the problem of gradient vanishing caused by increasing depth in the neural network. The residual network block $H(\cdot)$ can be expressed as follows:

$$H(\mathbf{x}) = F(\mathbf{x}, \Theta) + \mathbf{x} \quad (4)$$

where $F(\cdot, \Theta)$ is the "convolution-pooling" structure in the residual network block and Θ is the parameter. Consequently, for the L -layer residual network blocks, the output $\mathbf{z}$ can be expressed as follows:

$$\mathbf{z} = (H_1 \circ \dots \circ H_L)(\mathbf{x}_1) = \mathbf{x}_1 + \sum_{i=1}^L F_i(\mathbf{x}_i, \Theta_i) \quad (5)$$

where $\mathbf{x}_i$ is the output of the i -th residual network block. Finally, we flatten its output $\mathbf{z}$, enabling information extraction from the original input byte stream.

In a network flow, uplink and downlink loads are respectively generated by the client and the server, thereby often exhibiting different characteristics in textual domain. In order to avoid interference caused by the confusion of uplink and downlink loads, it is essential to separate them during feature extraction, and then perform feature fusion. For a network flow, we input its uplink load (B_{up}) and downlink load (B_{down}) into the network structure separately, so as to extract their textual domain features, respectively, that is, z_{up} and z_{down} . After concatenating the two output results, we input them into a Multi-Layer Perception (MLP) structure for feature fusion and dimensionality reduction, so as to obtain the final representation result of the network flow text domain information. Based on this representation result, the downstream identification task can be achieved by connecting a classifier at the output of the network structure.

Text-Net is trained through supervised learning. We apply cross-entropy as the loss function, which is defined as:

$$\mathcal{L}_{cls} = -\frac{1}{N} \sum_{i=1}^N \sum_{c \in \mathcal{C}} y_{ic} \log(p_{ic}) \quad (6)$$

where y_{ic} is the ground truth label, and p_{ic} is the predicted probability that the i -th sample belongs to category c .

D. PS-Net for Temporal Domain

In ExpMD, PS-Net is an identification model based on bidirectional multi-layer RNN structure. Through representation learning, PS-Net can extract temporal domain features from the packet sequence of the network flow. The detailed structure of PS-Net is shown in Fig. 3.

The input to PS-Net is the packet-length sequence of the network flow. The maximum packet length (*i.e.*, MTU) is set to 1500. To distinguish the direction of packets, the uplink packet length is set to positive while the downlink packet length is set to negative. Thus, the packet sequence is converted to an integer sequence in the interval $[-MTU, +MTU]$.

First, the packet-length sequence $L = [L_1, L_2, \dots, L_n]$ is converted into an embedding sequence $E = [e_1, e_2, \dots, e_n]$ through the embedding layer. We then apply the bidirectional multi-layer RNN structure to extract its sequence information. The low-level expressions learn the local features, while the high-level expressions are composed of combinations of low-level expressions to obtain global features. Formally, the input of the i -th layer at time step t is the output $o_t^{(i-1)}$ of the $(i-1)$ -th layer at time step t , that is,

$$\begin{aligned} \vec{h}_t^{(i)} &= \overrightarrow{\text{RNN}}\left(\vec{h}_{t-1}^{(i)}, o_t^{(i-1)}\right), \\ \overleftarrow{h}_t^{(i)} &= \overleftarrow{\text{RNN}}\left(\overleftarrow{h}_{t+1}^{(i)}, o_t^{(i-1)}\right), \end{aligned} \quad (7)$$

where $\vec{h}_t^{(i)}$ and $\overleftarrow{h}_t^{(i)}$ are the forward and backward hidden states, respectively.

For the first layer ($i = 1$), we should input the embedding sequence from both directions to the bi-RNN, that is, $o_t^{(0)} = e_t$. The output of the i -th layer bi-RNN is the concatenation of the hidden state of both directions, *i.e.*,

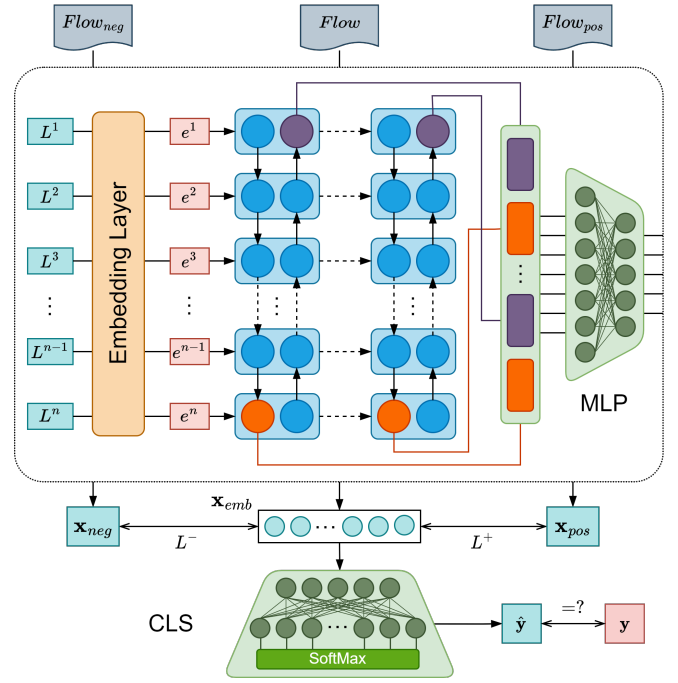


Fig. 3. The network structure of PS-Net.

$o_t^{(i)} = [\vec{h}_t^{(i)}, \overleftarrow{h}_t^{(i)}]$. After that, the hidden layers of each layer in both forward and backward directions of each bi-RNN structure are concatenated together to obtain the feature vector z of the input sequence as follows:

$$z = [\vec{h}_n^{(1)}, \overleftarrow{h}_1^{(1)}, \dots, \vec{h}_n^{(J)}, \overleftarrow{h}_1^{(J)}] \quad (8)$$

where J is the layer number of the bi-RNN structure.

With this setting, z contains bidirectional temporal information of the whole sequence, and the information fusion and dimensionality reduction are realized by feeding them into a MLP structure. Then, we obtain feature vectors that can represent the temporal domain information of the original packet-length sequence. The downstream identification task can be achieved by feeding them into a classifier.

In order to improve the training efficiency of PS-Net, we also introduce a contrastive learning in the training process besides classification task. Contrastive learning uses label similarity instead of original labels during training, leading to the model better suited for the data space. We apply the Triplet Network [31] to achieve contrastive learning. For an input sample x , a positive sample x^+ with the same category and a negative sample x^- with a different category are also randomly selected. PS-Net can respectively generate their representation vectors for these three samples (*i.e.*, $f(x)$, $f(x^+)$, and $f(x^-)$). In the low-dimensional representation space, it is required to keep the input sample close to the negative sample and far away from the negative sample as much as possible. The contrastive learning process is achieved

by minimizing the triplet loss, which is defined as:

$$\mathcal{L}_{triplet} = \max \{D(f(\mathbf{x}), f(\mathbf{x}^+)) - D(f(\mathbf{x}), f(\mathbf{x}^-)) + m, 0\}, \quad (9)$$

where m is called margin value. We apply the Euclidean distance to measure the similarity of the representation vectors:

$$D(f(\mathbf{x}_1), f(\mathbf{x}_2)) = \|\mathbf{f}(\mathbf{x}_1) - \mathbf{f}(\mathbf{x}_2)\|_2. \quad (10)$$

Finally, PS-Net is trained through multi-task learning. The loss function is the weighted sum of cross-entropy loss and triplet loss, that is:

$$\mathcal{L} = \mathcal{L}_{cls} + \alpha \mathcal{L}_{triplet}, \quad (11)$$

where α is the weight of loss function.

E. Ensemble Based on Soft Voting

Based on the identification results of the four single-domain models, we employ a soft voting mechanism to determine the category of the network flow. The voting value of each model corresponds to its identification probability. In contrast to hard voting, soft voting can mitigate the interference caused by low-confidence identification results from individual models in the final decision. Considering the different identification performance of models, it is more reasonable to set different voting weights for models of different feature-domain. Hence, the final voting result is the weighted sum of the identification probabilities from each model.

We set the model's weight in the feature domain $d \in \mathcal{D}$ as λ_d where $\mathcal{D}$ is the set of feature domains. For all the samples in the training set, $\mathbf{P} = [\mathbf{p}_d^1, \mathbf{p}_d^2, \dots, \mathbf{p}_d^N]$ consists of their output probability vectors in the feature domain d , and $\mathbf{Y} = [\mathbf{y}^1, \mathbf{y}^2, \dots, \mathbf{y}^N]$ is constructed by one-hot encoding of their label data. The problem of finding the optimal weights of models can be reformulated as a constrained optimization problem, which is expressed as follows:

$$\begin{cases} \max_{\lambda} \|\mathbf{Y} \odot \sum_{d \in \mathcal{D}} \lambda_d \mathbf{P}_d\|_1 \\ \text{s.t.} \quad \sum_{d \in \mathcal{D}} \lambda_d = 1, \quad 0 \leq \lambda_d \leq 1 \quad (d \in \mathcal{D}). \end{cases} \quad (12)$$

where $\odot$ is the Hadamard product of matrixes.

Therefore, according to the Maximum Likelihood Estimate (MLE), the final identification result can be expressed as:

$$\hat{y} = \arg \max_{c \in \mathcal{C}} \sum_{d \in \mathcal{D}} \lambda_d \mathbf{p}_d, \quad (13)$$

where $\mathbf{p}_d$ is the output probability of the unknown network flow in the feature domain d and $\mathcal{C}$ is the label set.

F. Explanation

Feature attribution serves as a primary method for enhancing the interpretability of identification models. It is also known as the instance-wise explainer, which provides an explanation for the prediction of the target model on any individual instance. In expMD, the models applied in different feature domains are different, thus their applicable explainable methods are also different. We apply SHapley Additive exPlanations (SHAP)

[32] and Integral Gradient (IG) algorithm [33] to achieve feature attribution in the framework.

1) *SHAP*: SHAP [32] provides a way to understand the contribution of individual features to model's predictions. The SHAP is better suited for explaining models based on tabular data, such as tree-based models. For tabular data, features typically possess clear semantics. SHAP can offer more intuitive feature importance explanations in such cases. Consequently, we employ SHAP for feature attribution on Meta-RF and BD-boost.

SHAP assigns each feature an importance score based on Shapley Values, enabling users to explain why a model made a specific prediction. For an input instance $\mathbf{x}$ with all features N , the Shapley Value of its feature i is defined as follows:

$$\phi_i(\mathbf{x}) = \frac{1}{|\Pi_i(N)|} \sum_{\pi \in \Pi_i(N)} [f(\mathbf{x}_{\pi \cup \{i\}}) - f(\mathbf{x}_{\pi})] \quad (14)$$

where $\Pi_i(N) = \{S | S \subseteq N \setminus \{i\}\}$ is the set of subsets for feature set $N \setminus \{i\}$.

Shapley Value is the attributions method that satisfies the properties efficiency, symmetry, dummy and additivity, which together can be considered a definition of a fair payout. But the Shapley Value requires a high computational complexity. Many methods, such as TreeSHAP [34], are proposed to estimate the Shapley Value in a fast speed.

2) *Integrated Gradients (IG)*: The IG algorithm [33] is a technique for explaining the predictions made by models, particularly Deep Neural Networks. It first computes the gradient of the model's output $F(\cdot)$ with respect to the feature i along a path from a baseline input $\mathbf{x}'_i$ (usually a reference or zero-input) to the actual input $\mathbf{x}_i$. By integrating these gradients along the path, it assigns the importance score of feature i as follows:

$$\phi_i(\mathbf{x}) = (\mathbf{x}_i - \mathbf{x}'_i) \int_0^1 \frac{\partial}{\partial \mathbf{x}'_i} F(\mathbf{x}'_i + \alpha(\mathbf{x}_i - \mathbf{x}'_i)) d\alpha. \quad (15)$$

For representation learning models based on Neural Networks, they typically operate on continuous input data directly, such as text and images. IG captures complex feature interactions effectively through calculating gradients by integrating along a path in the input space, making it more suitable for representation learning models. Consequently, we employ IG for feature attribution on Text-Net and PS-Net.

V. EVALUATION

In this section, we conduct a set of experiments¹ to evaluate the performance of proposed ExpMD framework. Our experiments were simulated on a server equipped with Intel(R) Xeon(R) Gold 5222 CPU @ 3.80GHz, NVIDIA GeForce GTX 3090 GPU, and 128GB of RAM.

The datasets used in this paper are constructed based on the public dataset NUDT_MobileTraffic [35]. We selected 104 application from this dataset and subsequently created two datasets, one for training and another for testing.

¹The experimental source code is available at <https://github.com/Shangshu-LAB/ExpMD>

A. Identification Accuracy

In the ensemble process, the voting weights are optimized based on Eq. 13. The voting weights of different sub-models are illustrated in Table I. Among them, Meta-RF has the largest weight, while PS-Net has the smallest weight. This indicates that the metadata domain contains the most important information for describing network flows. Both the BD domain and the textual domain are related to the payload of the network flow, while the temporal domain is related to the packet transmission process of the network flow. According to the voting weights, for network flows, the payload information is more important than packet sequence information.

TABLE I
THE VOTING WEIGHT OF DIFFERENT MODELS.

Model	Meta-RF	BD-boost	Text-Net	PS-Net
Weight	$\lambda_{meta}=0.38$	$\lambda_{BD}=0.24$	$\lambda_{text}=0.27$	$\lambda_{ps}=0.11$

The detection results of the ensemble model and the single-domain model is shown in Table II. The multi-domain ensemble method we proposed can achieve an accuracy of 0.9755, which is a good identification performance. In addition, single-domain models are also able to achieve more than 90% accuracy. However, some classification errors that occur on single-feature-domain models can be corrected by models of other feature domain. It demonstrated that the multi-domain ensemble method can effectively improve the network traffic identification effect.

TABLE II
THE STATISTICS OF THE IDENTIFICATION RESULTS FOR EXPMD AND EACH OF ITS SUB-MODELS.

Model	Meta-RF	BD-boost	Text-Net	PS-Net	ExpMD
Acc macro- F_1	.8927 .8921	.9103 .9111	.9508 .9517	.9053 .9055	.9755 .9756
F_1^*	[0.98, 1]	5 (4.8)	12 (11.5)	33 (31.7)	8 (7.7)
	[0.95, 0.98)	20 (19.2)	25 (24.0)	33 (31.7)	25 (24.0)
	[0.9, 0.95)	30 (28.8)	35 (33.7)	24 (23.1)	36 (34.6)
	[0.8, 0.9)	37 (35.6)	21 (20.2)	12 (11.6)	24 (23.1)
	[0.65, 0.8)	12 (11.5)	11 (10.6)	2 (1.9)	11 (10.6)

* The values in parentheses are the percentages whose F_1 falls in the specified range.

After that, we performed a statistical analysis of the F_1 score of the single-category identification result. For 98% of applications, the proposed multi-domain ensemble method can achieve the identification effect of F_1 scores above 0.9. Among them, there are 55.7% of applications that can achieve an F_1 score can reach 0.98, which is an excellent identification effect. In terms of the overall metrics, among the four single-feature-domain models, the identification effect of Text-Net is the closest to that of the ensemble method ($\Delta_{macro-F_1} = 0.0239$). However, it can only achieve an F_1 higher than 0.98 recognition effect for 31.7% of applications. In contrast, the integration method significantly increases the number of apps that can be identified with high accuracy.

B. Comparison Experiments

To obtain a comprehensive understanding of the performance of our method, we compare ExpMD with several existing state-of-the-art (SOTA) methods.

For Metadata and BD domain, their features are manually extracted statistical features, which are more suited for traditional machine-learning-based methods. Therefore, we employ traditional machine learning algorithms including Naive Bayes (NB), k -Nearest Neighbors (k -NN), Support Vector Machine (SVM), MLP, C4.5, RF, and XGBoost as baselines. For textual Domain, we employ Wang *et al.*'s method (W) [7], [13], Zhang *et al.*'s method (Zh) [10], and DeepPacket [8] as baselines, which utilize the first 784 bytes of payload from network flows. For temporal domain, we employ AppScanner [2], [3], FlowLens [5], and ETC-PS [6] as base lines, whose network flow information utilized is packet length sequences.

The comparison results with several existing methods are illustrated in Table III. The experimental results demonstrate that our proposed multi-domain ensemble method is able to significantly enhance the identification performance compared with the existing methods.

TABLE III
THE COMPARISON BETWEEN EXPMD AND EACH OF ITS SUB-MODELS WITH EXISTING SOTA METHODS.

Feature Doamin	Model	Metrics		
		Acc	macro- F_1	$F_1 > .95^*$
Metadata Domain	NB	.3198	.2765	0(0.0)
	k -NN	.5878	.5815	0(0.0)
	SVM	.0839	.0574	0(0.0)
	MLP	.3362	.2998	0(0.0)
	C4.5	.8693	.8683	14(13.5)
	XGBoost	.8476	.8484	20(19.2)
	Meta-RF	.8927	.8921	25 (24.0)
Byte-Distribution (BD) Domain	NB	.3198	.2765	0(0.0)
	k -NN	.4774	.4854	3(2.9)
	SVM	.7240	.7377	9(8.7)
	MLP	.7686	.7694	8(7.7)
	C4.5	.8382	.8366	11(10.6)
	RF	.8944	.8964	27(26.0)
	BD-boost	.9103	.9111	37 (35.6)
Textual Domain	W-1dCNN	.7174	.681	13(12.5)
	W-2dCNN	.6368	.5856	5(4.8)
	Zh-1dCNN	.8459	.8185	22(21.2)
	Zh-2dCNN	.7951	.7586	15(14.4)
	Zh-3dCNN	.4884	.4217	1(1.0)
	Zh-MLP	.6202	.569	6(5.8)
	DP-CNN	.7057	.6761	12(11.5)
	Text-Net	.9508	.9517	66 (63.5)
Temporal Domain	AppScanner	.9015	.9011	21(20.2)
	ETC-PS	.8890	.8885	18(17.3)
	FlowLens	.8719	.8712	12(11.5)
	PS-Net	.9053	.9055	33 (31.7)
	ExpMD	.9755	.9756	93 (89.4)

* The values in parentheses are the percentages with F_1 exceeding 0.95.

Then, we compare existing methods with the designed models in each feature domain.

1) *Metadata and BD Domain:* The metadata and BD domains primarily consist of statistical features. Thus, we compare several traditional-machine-learning-based models with Meta-RF and BD-boost. As shown in Table III, the Meta-RF model based on Random Forest algorithm performs the best in metadata domain, while the BD-boost model utilizing XGBoost algorithm demonstrates the best performance in the BD domain. Compared to other machine learning algorithms, tree-based models generally show better performance on tabular data dominated by statistics. In addition, Random Forest and XGBoost show significant advantages of ensemble learning in contrast with decision tree algorithm.

2) *Textual Domain:* As illustrated in Table III, the identification performance of Text-Net significantly better than existing methods in the textual domain. The Text-Net employs byte embedding technique and separates the uplink and downlink loads in the feature extraction process, which are beneficial for learning the relationships between bytes within the load. In addition, the multi-layer residual network structure in Text-Net can also improve its representation learning ability. Hence, Text-Net is capable of automatically extracting text-domain features from network flow loads, which contribute to identification tasks.

Table III also demonstrates that the 1dCNN-based models are superior to other methods. This is due to the fact that the load of network flow is one-dimensional textual data, making them better suited for a 1d-CNN architecture. It should be noticed that the process of hard-cutting and rearranging network flow loads into two-dimensional or three-dimensional data will disrupt the logical relationships between bytes within the original load, which introduced interference for representation learning.

3) *Temporal Domain:* From Table III, it is evident that PS-Net outperforms existing methods based on temporal domain in the identification task. Compared with these methods based on manual feature extraction, the PS-Net we designed can automatically extract more efficient features from packet sequences through representation learning.

C. Explanation Study

We selected a TLS-encrypted network flow generated by an application called Luckin Coffee as an instance to identify and perform feature attribution. The results of identification and feature attribution are illustrated in Fig. 4. In the four feature domains, the probabilities for identifying this network flow as the Luckin Coffee application are 0.7600 (Meta-RF), 0.9644 (BD-boost), 1.0000 (Text-Net), and 1.0000 (PS-Net), respectively. Therefore, our ensemble method ultimately determines that this network traffic belongs to the Luckin Coffee application, which is reasonable.

Then, we performed feature attribution based on this identification result in four feature domains. Fig. 4(a) illustrates the feature attribution results of the network flow in the metadata domain. It can be observed that all features in the metadata domain support this identification result. Fig. 4(b) presents the feature attribution results of the network flow

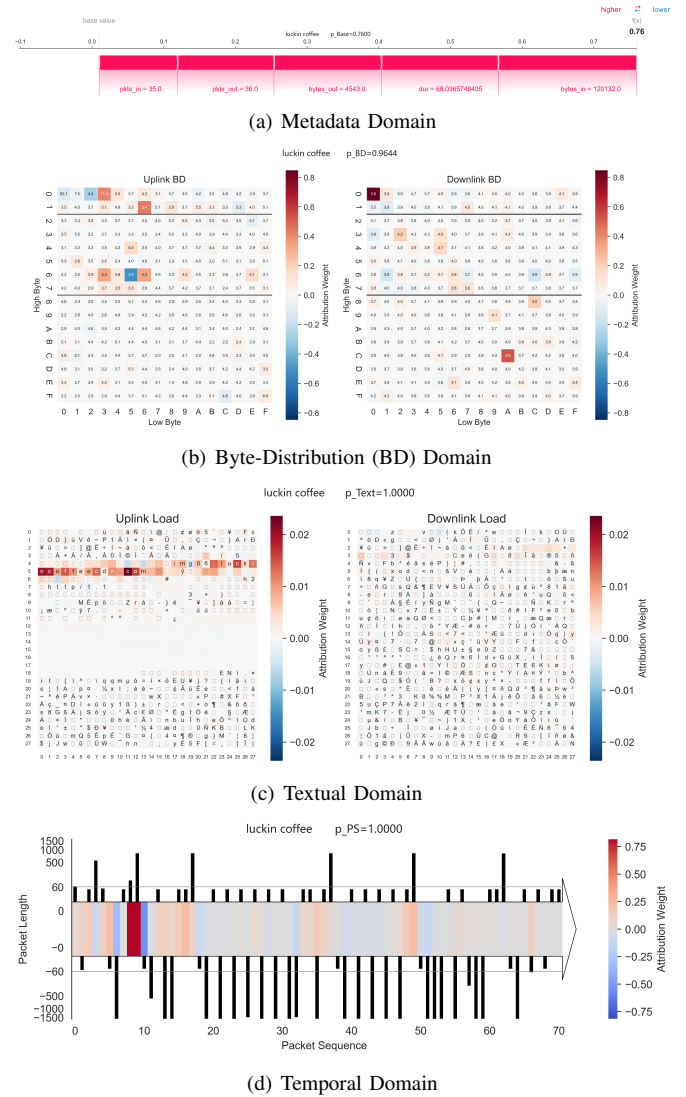


Fig. 4. The explanation of the identification result for a network flow sample by ExpMD in four different feature domains.

in the Byte-Distribution domain. Because the network flow has been encrypted by TLS, most of the bytes in the payload are distributed uniformly except for a few specific bytes such as 0×00 . As shown in Fig. 4(b), only a few bytes have a significant impact on the identification result, while the majority of bytes have negligible influence on the identification result. For this network flow, the distribution of 0×00 in the downlink load contributes the most to the identification result. Besides, bytes such as $0 \times CA$ in the downlink load and 0×03 , 0×16 , 0×63 , and 0×66 in the uplink load also notably support the identification result. On the contrary, the distribution of 0×65 in the uplink load has the most significant adverse effect on the identification result.

The feature attribution results of this network flow in the textual domain are displayed in Fig. 4(c). Because of the encryption technology, most of the textual information in its payload lacks plaintext meaning. Consequently, the textual

information for most parts of the payload does not affect the identification results. However, the plaintext information exchanged during the initial handshake process of the TLS flow can be utilized for identification. As shown in Fig. 4(c), the string `img06.luckincoffeedn.com` appearing in the uplink load has a higher attribution weight in the textual domain, which is strongly associated with the Luckin Coffee application. According to the TLS protocol format, this string is the Server Name Indication (SNI) from the `ClientHello` message during the TLS handshake process, which is applied to indicate the host name (domain name) of the target server with which the client wants to establish an encrypted connection. It demonstrates that Text-Net has the ability to identify plaintext information within encrypted network flows that contributes to identification, and automatically extract features.

On the temporal domain, Fig. 4(d) displays the feature attribution results of the network flow for packet length sequences. Based on the attribution weights, it can be observed that PS-Net successfully identifies specific patterns of the packet length sequences, which contribute to the identification. In addition, it is worth noting that the first 20% of the packet sequence has a more significant impact on the identification results as illustrated in Fig. 4(d).

VI. DISCUSSION

In ExpMD, network flow features can be categorized into four feature domains: metadata domain, byte-distribution domain, textual domain, and temporal domain. By constructing and training models in single domain, the information of network flow in different feature domains can be fully extracted. Subsequently, we ensemble the optimal models from each feature domain to attain superior identification performance.

In a network flow, the uplink flow and downlink flow have different functions. Especially in the textual domain and byte-distribution domain, there exist differences in the information contained in two directions of flows. Therefore, in the feature extraction phase, we segregate uplink and downlink flows and subsequently merge their features during the identification phase. This process is in the design of both BD-boost and Text-Net. As for the model design, we employed byte embedding in Text-Net to mitigate the influence of byte values on their actual meaning. We introduced contrastive learning in PS-Net to lead the model better suited for the data space, which enhances its representation capacity. These aforementioned designs enable our framework to achieve better identification of network flows. Additionally, we employed explainable methods in our framework to attribute features, thereby enhancing the interpretability of the identification results.

Next, we will discuss the roles played by different feature domains in the identification of network flows. The experimental results clearly demonstrate the significant role played by the textual domain, which is easily explainable. In manual traffic analysis, when certain feature fields (e.g., domain names) appear in packets, analysts can typically accurately determine the functionality of network flows. Text-Net can effectively capture these feature fields in network flow payloads, thereby

achieving a high identification accuracy in the textual domain. For encrypted traffic, the effective textual domain information is limited to the plaintext information transmitted during the initial handshake process, such as the Server Name Indication (SNI) in TLS flows. This information is sufficient to achieve accurate identification. However, deliberate removal or obfuscation of these information will significantly diminish the role of textual domain information in identification tasks. The information of byte-distribution domain can reflect the overall characteristics of the byte stream, such as the file type of the payload. As for encrypted network flows, the bytes within their payloads exhibit a uniform distribution. Consequently, the role of byte-distribution domain information is limited in the identification of encrypted traffic. The metadata domain and temporal domain primarily capture the behavioral information of network flows. Behavioral characteristics analysis constitutes the primary approach for analyzing encrypted traffic. In network traffic analysis, the likelihood of misclassification is higher in behavioral identification compared to feature field identification in textual domain. It is because not only can various categories of network flows exhibit similar network behaviors, but also there can be significant differences in behavior among network flows within the same category.

VII. CONCLUSION

Network traffic identification has a significant impact on the QoS of network. But underutilization of information and weak of interpretability bring great challenges to the performance of existing approaches. In this paper, we present a universal and explainable framework for network traffic identification, named ExpMD. It can make full use the multi-domain information of network flows and achieve feature attribution for identification results. As a result, it is able to mitigate the underutilization of information and the lack of interpretability for identification models, effectively. Extensive experiments demonstrated that our multi-domain ensemble method can effectively improve the network traffic identification effect. Each sub-model performed at a state-of-the-art effect within their feature domains, respectively. Furthermore, our framework can also generate high-fidelity, sparse, complete, and stable explanation results for network flows.

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